

Total No. of Questions : 6]

SEAT No. :

P11

[Total No. of Pages : 2

TE/INSEM/APR-14
T.E. (Civil) (Semester - II)
301010-STRUCTURAL DESIGN - II
(2015 Pattern)

Time : 1½ Hours]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6*
- 2) *Figures to the right indicate full marks.*
- 3) *Use of IS 456-2000 and non programmable calculator is allowed.*
- 4) *Neat diagrams must be drawn wherever necessary.*
- 5) *Mere reproduction from IS Code as answer, will not be given full credit.*
- 6) *Assume any other data, if necessary.*

Q1) a) Explain the meaning of balanced section with respect to WSM & LSM.[3]

- b) A beam of size 230 mm × 600 mm with effective cover of 50mm is simply supported over a span of Span 5m. The reinforcement consists of 4 bars of 20mm diameter at tension face. Find safe intensity of uniformly distributed load that can be placed on beam. Use WSM. Materials are M25 & Fe415. [7]

OR

Q2) a) Explain need of under reinforced section in LSM design. [3]

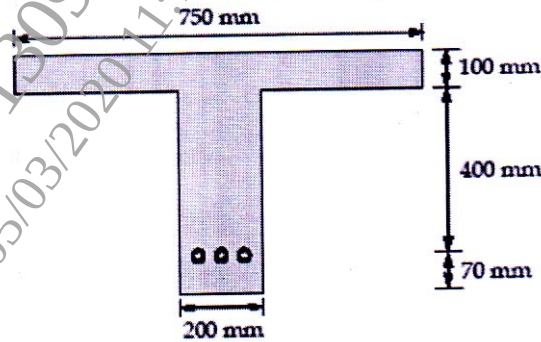
- b) Using stress and strain diagrams for a balanced singly reinforced section as per WSM, derive the design constants (k = Neutral axis constant, j = Lever arm constant, and Q = Moment of resistance constant). Use M20 grade of concrete and Fe 415 grade steel. [7]

Q3) a) Determine minimum strain at yielding in tension steel having yield stress 500 MPa and Young's modulus $E_s = 210$ GPa, as per LSM. [2]

P.T.O.

- b) A T-beam of span 6m has cross-section as shown in the figure. It consists of 3-#16 bars as shown. Calculate ultimate moment, of resistance and find the safe superimposed load that can be carried by the beam. Use M20 concrete and Fe415 reinforced steel. [8]

(All dimension are in mm)



OR

- Q4)** Design a RC slab for a store room having clear dimensions as 4.8m × 3.8m. The slab is supported by 230mm wide beam along all the four edges with corners of slab held down. The slab carries live load of 3 kN/m² & floor finish of 1.5 kN/m². Use M25 grade of concrete and Fe 500 grade of steel. Also show details of reinforcement. Use LSM. Neglect check for shear. [10]

- Q5)** A simply supported slab of clear dimensions 3.2 m × 8.4m is to be provided for a corridor. The slab carries live load of 4 kN/m² & floor finish of 2.0 kN/m². Design the slab using M20 grade of concrete and Fe500 grade of steel. Draw neat sketches showing details of reinforcement. Use LSM [10]

OR

- Q6)** Design the second flight (midlanding level to first floor level) of a dog legged staircase of public building with the following data : [10]

- Floor to floor height = 3.6m
- Rise = 150mm; Tread = 300mm; Width of flight = 1.5m
- Width of mid level landing = 1.5 m
- Width of floor level landing = 2.0m
- Width of supporting beams = 300mm
- Supporting beams are provided at the outer edges of both landings
- Material = M30, Fe 500
- Draw details of reinforcement

Use LSM approach. Neglect check for shear.

